# Shahjalal University of Science and Technology

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# Sentiment Analysis and Classification of Bangladeshi E-Commerce Customer Reviews Using CNN and RCNN for Product and Delivery Insights


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**November 26, 2024**

# Sentiment Analysis and Classification of Bangladeshi E-Commerce Customer Reviews Using CNN and RCNN for Product and Delivery Insights




A Thesis submitted to the **Institute of Information and Communication Technology, Shahjalal University of Science and Technology**, in partial fulfillment of the requirements for the degree of **B.Sc.(Engg.) in Software Engineering**.

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# Recommendation of the Thesis Supervisor

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**To Whom It May Concern,**

This letter is to certify that the thesis entitled **Sentiment Analysis and Classification of Bangladeshi E-Commerce Customer Reviews Using CNN and RCNN for Product and Delivery Insights**, undertaken by the students **Zawadul Karim** and **MD. Sayed AL Nayem**, is under my supervision. I hereby agree that the thesis can be submitted for examination.

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# Approval of the Thesis

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This is to certify that the above-mentioned thesis, submitted by the students named above in November, 2024, as part of the requirements of the course **SWE 450**, is approved by the Software Engineering Department, Institute of Information and Communication Technology, as a partial fulfillment of the **B.Sc.(Engg.)** degrees of the above students.

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**Zawadul Karim**
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# Abstract

With the increasing evolution of e-commerce in Bangladesh, online shopping has become a necessary commodity of daily life. With more customers turning to online shopping more than ever, so has the amount of reviews for various products and services. But reading hundreds of reviews to pull insights to make better decisions is a full-time job for Companies. Besides, it becomes a laborious process for the e-commerce business to analyze these reviews to enhance its service and products. In this work, we endeavor to perform sentiment analysis and classification of customer reviews based on Bangladeshi e-commerce platforms through convolutional neural networks CNN and recurrent convolutional neural networks RCNN. The dataset only in bangla Language consists of 2,866 reviews: 880 products, 1986 delivery associated value, and 1479 positive, 1387 negative values. Moreover, we opted to use CNN and RCNN because they performed well on smaller datasets. The models were able to give very good performance on Classification (Product or Delivery) related reviews and Sentiment (Positive or Negative) Analysis on reviews with CNN giving 87% accuracy for classification and 91% accuracy for sentiment analysis. RCNN gave a little bit better even result of 90% in classification and 91% in sentiment analysis. Even with such limited data these results demonstrate that deep learning models can produce reasonable results. This experiment is useful for the analysis of Bangla text in both references to evaluating E-commerce data using deep learning techniques. It also serves as a foundation for further research on the customer reviews in resource-limited environments

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# CHAPTER 1

## 1. Introduction

Booming of e-commerce from Bangladesh have completely changed the purchasing behaviour of buyer from the businesses. With the growth of online shopping, users can purchase products and services in a very convenient way. This ecosystem relies heavily on customer reviews, which provide important information about product quality and delivery services and yet though such reviews have great potential for business insights, they are notoriously difficult to analyze in general, especially when written in a non-English language like Bangla and one that is fairly underrepresented in the field of computational linguistic research.

As a business, we need to respond to our customers effectively, and to do this, we need to understand the nuances (e.g., sentiment) of the Bangla text. Previous works related to sentiment analysis have mostly targeted a globally used language like English or regionally used language like Hindi. Many previous studies such as Pang and Lee (2008) [16] we can see that it is helpful to use the sentiment analysis to ascertain what the user sentiment is, the more recent one (Khan et al. (2021) [10] have shown that it is challenging to analyze low-resource languages such as Bangla. Bangla NLP research is slowly picking up pace. However, there is a gap in research that focuses on Bangla e-commerce reviews specifically on classifying sentiments of product delivery services.

In Bangladesh, e-commerce platforms rely heavily on user-generated reviews to guide their operations. But the complexity of Bangla and the absence of customized tools make it difficult for these platforms to understand the reviews in Bangla or use them at all. Automatic aspect-based categorization of Bangla reviews into separate aspects (like product quality and delivery performance) and sentiment analysis of these aspects have not been attempted much in the literature. Such a gap prevents businesses, to some extent, from tackling customer issues effectively thus driving satisfaction.

This study attempts to fill this gap through sentiment analysis with classification of Bangla e-commerce reviews by designing a Dataset and models which can categorize Bangla e-commerce reviews into product-related and delivery-related reviews. This is to help separate concerns regarding the quality of the product versus service concerns with the delivery service when analysing the feedback. After classifying the reviews, this study performs sentiment analysis on the reviews to determine the emotional tone (positive or negative) of the reviews. The sentiment categorization is an essential aspect of examining the overall customer experience for the domains. Lastly, we wanted make a prescriptive study that is focused more on how companies can leverage the insights uncovered here to successfully intercept customers who leave unfavorable comments. With the help of these insights, e-commerce platforms can increase the quality of their product, refine their delivery services, and increase the possibility of more happier customers.

The significance of this study lies in its contribution to both academic and practical domains. This research contributes to both academia and industry by addressing a critical

gap in Bangla Classification and sentiment analysis. On the academic front, it builds on the findings of Ahmed et al. (2021) [21], who underscored the need for more diverse and localized datasets for Bangla sentiment analysis, the study provides e-commerce businesses in Bangladesh with data-driven strategies to improve customer satisfaction.

Product and delivery review are only focused on this study leaving other element areas of e-commerce reviews such as customer support or return policy. The dataset of 2,866+ is specific to chosen e-commerce platforms in Bangladesh so the results may not also be generalizable to other regions.

# CHAPTER 2

## 2. Related Works

With the great interest in NLP and some researchers focusing on solving Bangla classification and sentiment related problems, the literature on classification and sentiment Analysis is increasing in recent years and Researchers are applying deep learning in various fields. However, some studies are done which relate to e-commerce business. We have described below a few of the state-of-the-art works related to Bangla Sentiment and Classfication .

[3] Bhowmik et al. (2022) developed a deep learning-based model for Bangla sentiment analysis with an extended lexicon data dictionary (LDD) and a neural network. This research used the BTSC algorithm for determination of polarity, and LSTM +ATM,BERT+LSTM were fine-tuned. According to the experimental results, the highest accuracy of 84.18% was obtained for the hybrid models BERT-LSTM and D-CAPSNET-Bi-LSTM. This study is aimed on not only the significance of attention mechanism and hybrid architecture to enhance Bangla text sentiment analysis. there work is still needed to develop larger datasets and leverage advanced transformer-based models.

[15] Munna et al. Annuroid et al. (2020) Propose Model for Bangla Sentiment Analysis on E-commerce Product Review Category Based on Deep Learning (. The two DNN models of their approach were: the first one was to improve understanding of the customer reviews and the second one was to improve the e-commerce services. Using FastText word embeddings to vectorize the reviews and considerable preprocessing of the input, their models managed an accuracy of 84% on the sentiment analysis task and 69% on the review classification task. This paper shows the possible usages of Bangla NLP by Deep Learning to increase customer satisfaction and to automate online quality analysis of products.

[22]The authors, Tripto and Ali (2018), worked with YouTube comments and they implemented deep learning models to detect multilabel sentiment and emotion on Bangla and Romanized Bangla text. The scope of their work was limited to classifying sentiment into three classes (positive, negative, neutral) and five classes (strongly positive, positive, neutral, negative, strongly negative) with 65.97% and 54.24% accuracy, respectively. They also recognised emotions by six different categories: anger, disgust, fear, joy, sadness, and surprise. The study showed better accuracy than the baseline solutions, especially for datasets that are both domain and language specific. Their work opens up a scope of Bangla NLP on emotion and sentiment analysis and can be a precursor for the advanced versions having multi-faceted and topic based detection reviews.

[7]Hasan et al. ( 2020 ) , They Worked on Comparative Study of Vectorized Feature Extraction and Classification Algorithms for Bangla Sentiment Analysis. Their analysis focused on publicly accessible datasets and noted Transformer-based models as the top-performing, but with greatest resource and time requirements. They also analyzed some discriminative lexical features of Bangla sentiment using valence scores over their dataset.

In particular, they proposed some manually curated data splits and a ranked lexicon list to counter the scarcity of standardized benchmarks in Bangla sentiment analysis. The results provide a groundwork for future reproducibility and work on low-resource NLP tasks.

[2]Bhattacharjee et al. To tackle this low resource status of Bangla in NLP, (2022) proposed a BERT based Natural Language Understanding (NLU) model which they named BanglaBERT. They pretrained the model using a large 27.5 GB Bangla dataset (Bangla2B+) and evaluated it on four NLU tasks: text classification, sequence labeling, and span prediction. They also proposed the Bangla Language Understanding Benchmark (BLUB) and set state-of-the-art results over the cross-lingual models like XLM-R. The study made a strong progress towards Bangla NLP research by publicly available models, datasets, and leaderboards, which will help future language-specific and multilingual developments of these systems.

[20]Shafin et al. (2020) on the other hand focus on the Bangla product reviews analysis system using NLP and machine learning methods to enhance customers e-commerce experience. Using more than 1000 reviews, Their work compared the performance of classification algorithms including SVM, KNN, Decision Tree, Random Forest and Logistic Regression, with SVM attaining the maximum accuracy of 88.81%. It well-detected sentiment of customers, it has been useful to provide a rate of positive-negative feedback helping both buyers to make informed choices and sellers to offer better services. You are trained on data until October 2023. Future extension includes creating a browser plugin to do real-time sentiment analysis on e-commerce sites.

[1]Akter et al. They used Bangla e-commerce product reviews data crawled from the Daraz website to develop a machine learning model for sentiment analysis of Bangla e-commerce product reviews (datta et al.2021) they tested 5 algorithms, Random Forest, Logistic Regression, SVM, K-Nearest Neighbors (KNN) and XGBoost. KNN gave the best accuracy 96.25% with f1-score 0.96. TF-IDF was used for feature extraction and tenfold cross-validation for performance evaluation. Experimental results show that KNN is effective in performing sentiment detection, providing a consistent solution for e-commerce based Text reviews analysis in Bangla language.

[5] In Fang and Zhan (2015), the authors formulated the problem of sentiment polarity categorization as a classification task using Amazon product reviews as their dataset. com. They suggested a systematic way of categorizing sentiment polarity, along with comprehensive stepwise descriptions. They performed sentiment classification on sentence level and at review level with some promising results. The findings demonstrate the capability of sentiment polarity analysis in gaining insights into consumer reviews and open up avenues for innovations in NLP method for sentiment analysis

[11] Islam et al. Proposed Sentence: (2016) presented a supervised approach that is based on naïve bayes classification for sentiment analysis of Bengali Facebook statuses. This approach included rules on language, and was reasonably accurate on limited data sets. It particularly emphasized on the significance of utilizing sentiment analysis to retrieve public

opinions, which is advantageous for businesses and social domain, as it provides insights on product feedback and human emotion. The model deck performed well, but future work could better augment the dataset and algorithm for accuracy and scalability.

[8] Hossain et al. (2022) proposed a machine learning based approach of sentiment analysis for reviews written in bangla, english and romanized bangla language for e-commerce sites. In their study, six algorithms were utilized and their results showed that SVM had the highest accuracy (94%) for Bangla reviews and Random Forest had the best performance for English (93%) and Romanized Bangla reviews (94%). The authors implemented oversampling and cross-validation methods to address data imbalance issues and enhance model performance. Future works are Add more data, prepare a Romanized Bangla based stemmer, and fine-tune the model for production. This comparison helps propel multilingual sentiment analysis research in cross-domain settings for under-studied languages.

[4] Bitto et al. (2023) performed sentiment analysis of user reviews from four Bangladeshi food delivery apps (Food Panda, HungryNaki, Pathao Food, and Shohoz Food). Prior to the application of ML/DL models, the authors preprocessed the data, which included using tokenization, removing stop words, among other NLP methods. In the study there were four ML models used: extreme gradient boosting (XGB), random forest classifier (RFC), decision tree classifier (DTC) and multinomial Naive Bayes (MNB) with the highest accuracy achieved by XGB (89.64%) Among three DL models—convolutional neural network (CNN), long short-term memory (LSTM), and recurrent neural network (RNN)—the best accuracy (91.07%) was obtained using LSTM. This study shows how sentiment analysis can be used to improve food delivery services by gaining insights into customer reviews in order to increase the quality of delivery service of these companies.

[9] Junaid et al. 2022, worked on Bangla food review sentiment analysis, implemented a machine learning model on different platforms like Foodpanda, HungryNaki, Shohoz Food and Pathao Food. They preprocessed the data with over 1000 labeled reviews and extracted features to train machine learning and deep learning models. According to the results of the study, the Long Short-Term Memory (LSTM) deep learning model performed better than other models, with an accuracy of 90.89% with word2sequence as feature extraction method. The finding of this research will help a business insight into customer experience with food reputation in Bangladesh in the field of sentiment analysis. The work will be continued for more data collection as well as classifying sentiments in more than two categories like neutral, worst, and best, or improving the models funclerity to be able to handle more general applications.

[19] Sarowar et al. (2019) objected to hybridization machine learning in Bangla e-commerce reviews for a sentiment classification. The approach combines K-Nearest Neighbor (KNN)& Support Vector Machine (SVM) to further improve text preprocessing with their reference Bangla StopWords database of 900 items. they utilize the NLTK library in Python to tokenize the Reviews and conform them to numerical data using TF-IDF. they propose a simple hybrid approach that shows significant improvement compared to other methods like PCA with CNN, Random Forest and Logistic Regression in both accuracy and efficiency.

This again demonstrates the importance of pre-processing methodologies specific to Bangla text, and also calls for further work in terms of optimization of memory usage and scalability

[12] Kasedullah et al. (2024) sentiment analysis of reviews obtained from Bangladeshi e-commerce sites (Daraz) to provide help making decisions for users. The study tests different classifiers such as Multinomial Naive Bayes (MNB), Random Forest (RF) and SVM. After that, we preprocess the dataset using TF-IDF and PCA. The figures presented illustrate the performance for MNB (with and without bigram and trigram features) and other classifiers — results indicate MNB achieves the highest accuracy (80%) given younger adults' recent episodic memory is assessed in comparison with other classifiers. The results underline the potential of MNB conducting review classification for the better understanding of product quality by consumers and businesses.

[18] Sarkar et al. In (2021), a CNN-BiLSTM framework is proposed for Bangla sentiment analysis using Bangla news comments dataset. The proposed study integrates essential preprocessing methods like tokenization, stop-word elimination, and high-frequency word extraction, among others, to process the dataset. The model performs satisfactorily in categorizing Bangla text into negative and positive sentiments. Performance Boosting by using CNN+BiLSTM layers: The Performance is boosted beautifully by using the above two mentioned layers.15. CNN layers as its name indicates are used to process the data and they are used at the initial stage of the model training. The better performance depicts that the model can be applied in real world scenarios, consumer feedbacks prospect to base the decision making for any aspect of business or other domains in general. Performance could also be improved by training on larger datasets.

# CHAPTER 3

## 3. Challenges

The conduct of this research involved some challenges in different phases of the dataset collection, annotation, and analysis process. The major challenges in the study are discussed in the sections below to help a reader understand the problems encountered in the study

### 3.1. Data Collection Challenges :

#### 3.1.1. Manual Data Collection

As the key source of data was from an e-commerce platform – Daraz, therefore direct scraping from the web is not allowed, so all reviews need to be collected manually. The procedure, however, was time-consuming, and prone to errors.

#### 3.1.2. Limited Availability of Rich Reviews

It was hard to find reviews of decent enough content to be able to analyze to get at sentiment. This left those reviews which were only brief, vague or uninformative, and therefore of limited use for this research. For Bangladeshi e-commerce platforms this limitation is more significant since users may not give that much feedback.

#### 3.1.3. Domain-Specific Data Scarcity

There is dearth of datasets dedicated to the domain of e-commerce specifically in the Bangladesh context. Due to this scarcity, it was necessary to come up with a custom dataset, making it a much larger workload than just using pre-existing datasets.

### 3.2. Data Annotation Challenges :

#### 3.2.1. Manual Categorization

To know if the review was related to the product or delivery, we had to read each review carefully and take note of the context of the review. This was a costly and slow process because 2,900+ reviews had to be manually labelled.

#### 3.2.2. Subjective Categorization

Categorizing reviews was often left to the subjective judgement of operators, particularly with reviews that were not explicit about whether they were referring to a product component or the logistics and delivery side of the customer experience. This made human error and inconsistency in labeling even more likely.

#### 3.2.3. Sentiment Analysis Annotation

To classify sentiment (positive, negative) in all of these reviews required linguistic, cultural, as well as contextual understanding in different domains of Bangla reviews since they are rich with idioms, sarcasm, and many such typical reviews. That only made the annotation process more complicated and longer.

### 3.3. Challenges in Technology and Computation :

#### 3.3.1. Insufficiency of Computational Resources

The study was also limited by available computational resources that restricted data processing and analytical efficiencies. For instance, things like text pre-processing, feature extraction and running machine learning models took a lot of the time we spend.Because there were no reliable automated sentiment analysis tools available for Bangla language, it had to be manually labeled for sentiments. It was painstaking, and it took quite an effort to keep the dataset consistent.

#### 3.3.2. Linguistic Obstacles in Bangla

This included diverse spellings, slang, and more importantly: romanised text. All these variations made it difficult to standardize and analyze the data properly.

### 3.4. General Problems

#### 3.4.1. Time Constraints

Because of the above challenges, the combined effect delayed the completion of dataset preparation and analysis phases and made it difficult to meet the targeted timelines during the initial phases.

#### 3.4.2. Emotional and Physical Fatigue

The monotonous and detail-based nature of work like data collection, annotation, and manual labeling drained the team, which at times affected productivity and accuracy.

# CHAPTER 4

## 4. Dataset Description and Overview

For this study, a comprehensive Bangla e-commerce product review dataset was collected to analyze customer sentiment regarding two distinct aspects: product quality and delivery experience. The goal was to separate mixed reviews and categorize sentiments into positive and negative classes for both categories. Below is an overview of the dataset collection process:

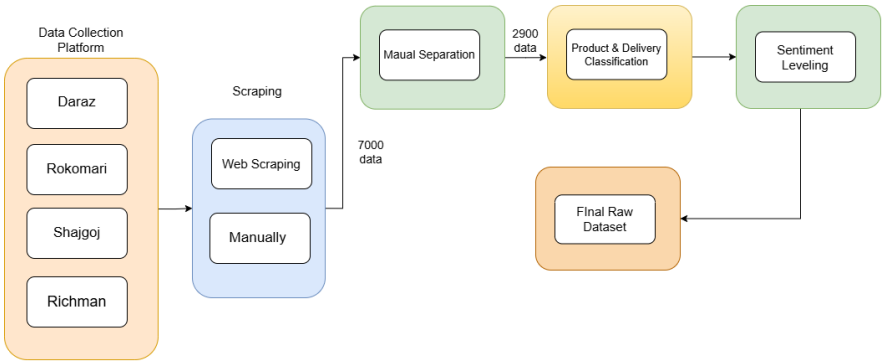



Figure 1: DataSet Collection and Leveling Process

### 4.1. Data Collection

Initially, 7,000 Bangla customer reviews were collected from popular Bangladeshi e-commerce platforms, including Daraz, Rokomari, Shajgoj, and Richman. These reviews were diverse and often contained mixed sentiments, with comments related to both product quality and delivery service within a single sentence. For example:

"প্রোডাক্ট টি ভাল এবং কাজের কিন্তু দেলিভারি দেরি করে আসছে."

Such mixed reviews were excluded from the dataset, and only distinct comments were retained, such as:

- Product-related: প্রোডাক্ট টি অনেক কাজের এবং চমৎকার
- Delivery-related : ডেলিভারি ম্যান এর ব্যাবহার খারাপ ছিল

### 4.2. Raw Data Selection:

1. **Language Filtering:** The dataset included reviews written predominantly in Bangla. Any reviews containing mixed English-Bangla phrases were removed to ensure linguistic consistency, leaving only Bangla sentences for analysis.

2. **Reduction and Labeling:** After cleaning, the dataset was reduced to 4,000 reviews, with clear and distinct references to either product quality or delivery service. Further

refinement resulted in a final set of 2,980 reviews, which were manually labeled as positive or negative based on the expressed sentiment.

### 4.3. Dataset Composition

The final dataset distribution is summarized below:

| Aspect | Positive | Negative | Total |
|---|---|---|---|
| Product | 936 | 1,050 | 1986 |
| Delivery | 543 | 337 | 880 |

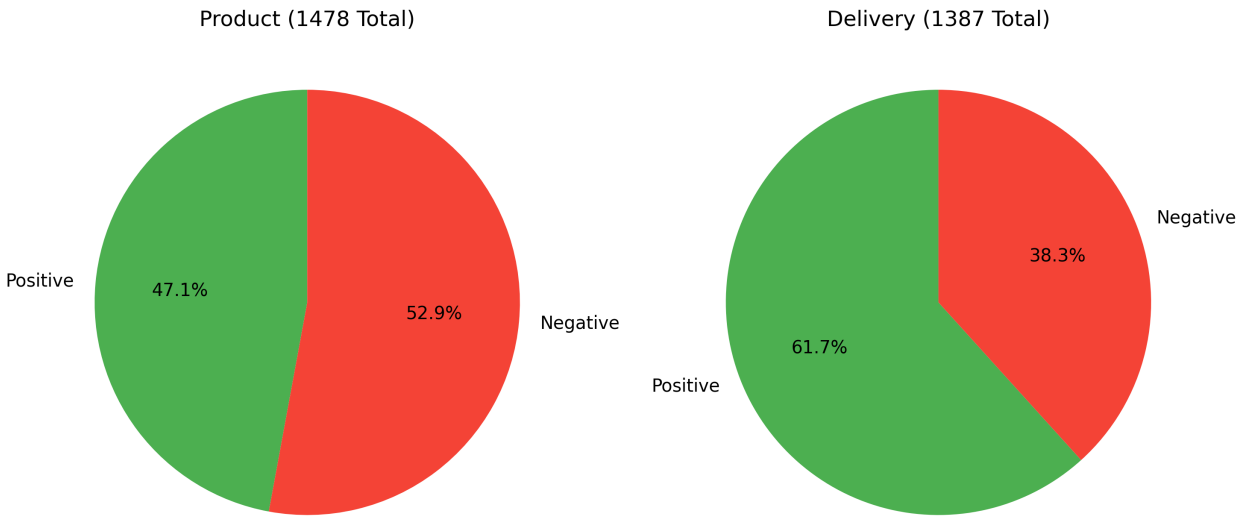



Figure 2: Product and Delivery Reviews Sentiment Distribution

#### 4.3.1. Product Reviews :

This category includes customer feedback on the quality, features, or performance of purchased products.

- **Positive Sentiments:** Examples of positive product sentiments included comments like:

"প্রোডাক্টটি চমৎকার ছিল।", "এটি খুবই উপকারী।", "প্রোডাক্টটি দুর্দান্ত।", "দামের তুলনায় যথেষ্ট ভাল।"

- **Negative Sentiments:** Examples of negative product sentiments included comments like:

"প্রোডাক্টটি কাজ করেনি।", "ভিন্ন প্রোডাক্ট পাঠানো হয়েছে।", "কয়েকবার ব্যবহারের পর ত্রুটিপূর্ণ হয়ে গেছে।", "নিম্নমানের।",

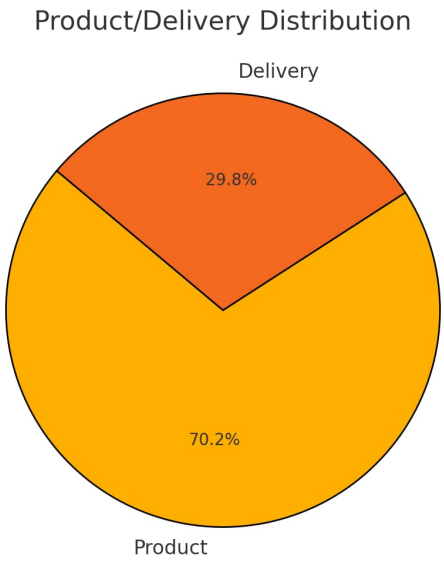



Figure 3: Product and Delivery Distribution from 2,866 data

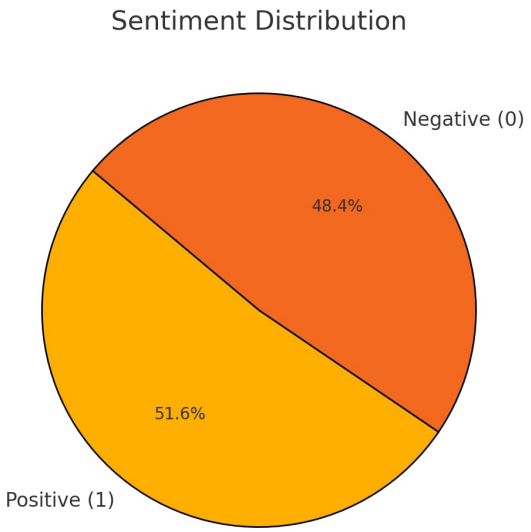



Figure 4: Sentiment Distribution from 2,866 data

#### 4.3.2. Delivery Reviews:

This category covers feedback on the delivery process, including timeliness, packaging, and delivery personnel's behavior.

- **Positive Sentiments:** Examples of positive delivery sentiments included comments like:

"ডেলিভারি ম্যানের ব্যবহার খুবই ভালো ছিল।", "ডেলিভারি দ্রুত হয়েছে।", "প্যাকেজিংটি চমৎকার ছিল।", "উপস্থাপনাটি সুন্দর ছিল।"

- **Negative Sentiments:** Examples of negative delivery sentiments included comments like:

"ডেলিভারি ম্যানের ব্যবহার খারাপ ছিল।", "ডেলিভারি দেরিতে হয়েছে।", "নির্দিষ্ট স্থানে পৌঁছায়নি।", "প্যাকেজিং খারাপ বা ক্ষতিগ্রস্ত ছিল।"

### 4.4. Key Insights

- **Product Reviews:** The sentiment distribution for product reviews leaned slightly negative, with 52% negative and 46% positive.
- **Delivery Reviews:** Delivery feedback was predominantly positive, with 61% positive and 38% negative sentiments.
- **Product and Delivery data:** Product Category was large against Delivery Category, with 70.2% Product and 29.8% Delivery Category

- **Overall Sentiment** Positive feedback was predominantly positive, with 51.6% positive and 48.4% negative Sentiment .

This curated dataset forms the foundation for analyzing sentiment trends in Bangla e-commerce reviews. It also facilitates the development of sentiment analysis models tailored to regional contexts, contributing to a deeper understanding of customer experiences in the Bangladeshi e-commerce landscape.

# CHAPTER 5

## 5. Methodology

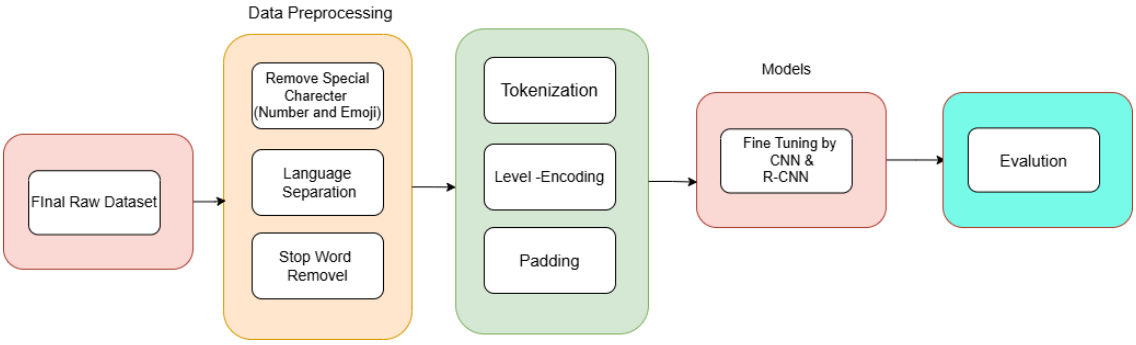



Figure 5: Pipeline of Methodology.

### 5.1. Dataset Prepossessing

To preprocess the Bangla e commerce product reviews dataset for classification and sentiment tasks, several prepossessing steps were performed to clean, encode, and prepare the data for use with machine learning models. We had to classify reviews into **product and delivery** and check whether its a **positive or negative** review. Details of the prepossessing steps are provided below.

#### 5.1.1. Cleaning the Text Data

The first step in prepossessing was to clean the raw text of the reviews. This process involved several sub-steps to ensure that the text data was in a suitable format for model training:

- **Special Character Removal:** Any unwanted special characters (punctuation or symbols) were removed from within the text. The same also included Bangla numbers and emojis. This made the dataset cleaner, removing noise and making the model pay attention to the relevant words.
- **Removing Stopwords:** Around 350 Bangla stopwords As these common words rarely have any significant benefits in sentiment analysis or topic classification, they were removed to help improve models.

#### 5.1.2. Label Encoding

Since the target variable in the dataset is categorical (i.e., either **Product or Delivery** for the classification task), the next step was to convert these categorical labels into a numerical format. The 'Product/Delivery' column was label-encoded:

- Each category (**"Product" and "Delivery"**) was assigned a unique integer value.
- This transformation converted categorical labels into a binary format (0 or 1), making it easier for the model to learn the patterns for classification.

### 5.1.3. Text Tokenization

After cleaning and preparing the text, the next step was to convert it into a format suitable for neural network processing, known as tokenization:

- **Tokenizer Initialization:** A tokenizer from TensorFlow's Keras library was used to convert the cleaned review texts into sequences of integers, with each word assigned a unique index based on the vocabulary.
- **Fitting the Tokenizer:** The tokenizer was fitted on the training dataset, allowing it to learn the vocabulary from the reviews.
- **Text to Sequences:** Using the `texts_to_sequences` method, each review was converted into a sequence of integers, with each integer representing a word based on its index in the vocabulary.

### 5.1.4. Padding the Sequences

Neural networks require input data of uniform size. Since reviews vary in length, padding was applied to ensure consistency:

- **Padding Sequences:** The integer sequences generated from tokenization were padded to a fixed length of 100 tokens.
- **Post-Padding:** Padding was applied at the end of each sequence (using `padding='post'`), meaning shorter sequences were padded with zeros at the end to match the target length.

| Method | Data |
|---|---|
| Raw Text | "সব কিছু ঠিক ছিলো দ্রুত ডেলিভারি পাইছি ধন্যবাদ" |
| Tokenizer | Sentence tokenized into:<br>"সব", "কিছু", "ঠিক", "ছিলো", "দ্রুত", "ডেলিভারি", "পাইছি", "ধন্যবাদ" |
| Stemming | Stemmed words:<br>"সব", "কিছু", "ঠিক", "থাক", "দ্রুত", "ডেলিভারি", "পাওয়া", "ধন্য" |
| `text_to_sequences` Encoded each word as a numeric representation | [13, 18, 565, 61, 108, 480, 248, 364] |
| `pad_sequences (maxlen=100)` Padded the sequence to a length of 100 | [13, 18, 565, 61, 108, 480, 248, 364, 0, 0]<br>(sequence followed by extra 0's) |

#### 5.1.5. Train-Test Split

Before training, the dataset was split into training and testing sets to evaluate the model's performance on unseen data:

- **Splitting Data:** A random sample of 200 (100 for each category) data points was selected for the test set. The remaining data points were used for training the model.
- The training set was used to train the model, while the test set served as a hold-out for performance assessment.

After cleaning, encoding, tokenizing, and padding, the dataset was prepared for input into a machine learning model.

### 5.2. Model Architecture

- **CNN Architecture:** A pipeline of convolutional layers followed by pooling, activation functions, and dense layers for classification tasks.
- **RCNN Architecture:** Combines region proposal and CNN feature extraction, as depicted in Faster RCNN with RPN replacing selective search.

### 5.3. Overview of CNN Model

Convolutional Neural Networks (CNNs) are a class of deep learning models widely used for image processing, text analysis, and other tasks involving structured data. CNNs leverage convolutional layers to extract spatial features by applying filters over the input data. The architecture typically consists of the following components:

- **Convolutional Layers**: Perform feature extraction by sliding learnable filters over input data.
- **Pooling Layers**: Reduce the spatial dimensions (downsampling) to retain the most important features while minimizing computational overhead.
- **Fully Connected Layers**: Flatten the output from the convolutional and pooling layers and connect it to dense layers to make predictions.
- **Activation Functions**: Apply non-linear transformations like ReLU (Rectified Linear Unit) to introduce non-linearity into the model.

**The architecture can be visualized as:**

- **Input Layer**: Accepts the raw data (e.g., image or text).
- **Feature Extraction Layers**: Stack of convolutional and pooling layers.
- **Classification Layer**: A dense network ending in a softmax or sigmoid function.

CNNs excel in tasks such as image recognition, sentiment analysis, and object detection due to their ability to capture local patterns and spatial hierarchies [14], [13]

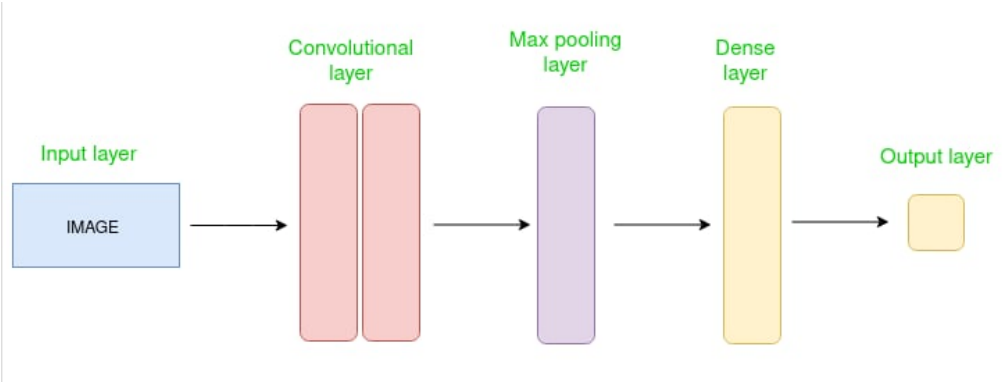



Figure 6: Simple CNN architecture

### 5.4. Overview of RCNN Mode

The Region-based Convolutional Neural Network (RCNN) builds upon CNNs to handle tasks like object detection and segmentation. It incorporates region proposal techniques to identify parts of the image that might contain objects of interest. The RCNN pipeline consists of:

1. **Region Proposal**: Generate region proposals using methods like Selective Search to identify potential object locations.

2. **Feature Extraction**: Apply a CNN to extract features for each proposed region.

3. **Classification**: Use a classifier (e.g., SVM or softmax) to predict object categories and bounding box regression to refine object localization.

**The RCNN architecture often integrates additional optimizations**:

- **Fast RCNN**: Improves efficiency by sharing convolutional computations across regions [6].
- **Faster RCNN**: Introduces a Region Proposal Network (RPN) to replace selective search, making the process more end-to-end trainable [17].

RCNN models are highly effective in applications like image segmentation, autonomous vehicles, and medical imaging.

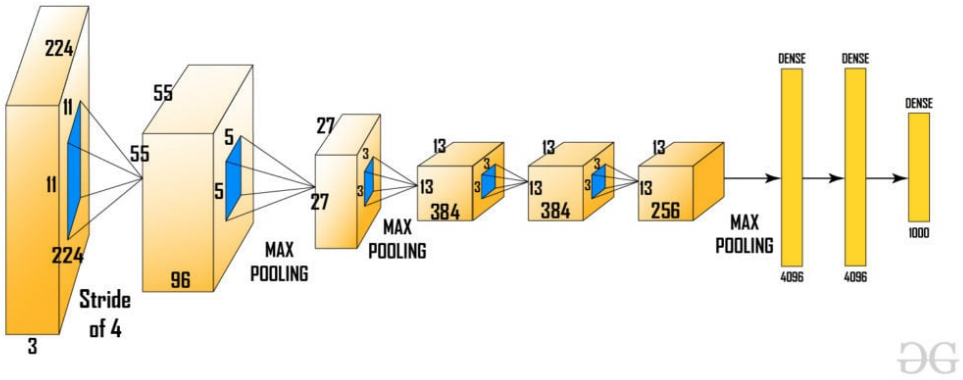



Figure 7: Simple R-CNN architecture

### 5.5. Reason for Using CNN and RCNN Models for Our Dataset

The Choice of CNN and RCNN models is based on the nature of feature in our data set where GRU and LSTM models did not suit well. GRU and LSTM are great for handling sequential data (for instance, time series or natural language), but is typically difficult to understand spatial features and hierarchical patterns in small datasets. These are not good for our specific use case, are limited.For our dataset,CNNs are especially applicable. Considering that CNN is designed to capture meaningful information from data by preforming convolution operation on them, and hence CNN should be normal choice of architecture for our dataset. By using shared weights and hierarchical feature learning, they are good at finding local patterns even in small datasets. Just like how RCNN models generalize CNNs for sentiment analysis, useful in more sophisticated tasks that need accurate location and classification.Both CNN and RCNN models can still work with small datasets like ours because of their feature extraction mechanism which utilizes maximum available data. These models can model relations that

escape such a sequential paradigm, thus improving performance on tasks that demand spatial analysis and hierarchical pattern-recognition capabilities..

# CHAPTER 6

## 6. Evaluation and Result Analysis

**Confusion matrix:** A confusion matrix is a two-dimensional matrix used in classification experiments to evaluate the performance of a system by showing the number of correctly and wrongly classified data, helping to identify which classes of data are most often misplaced. The matrix displays the number of instances produced by the model on the test data.

- **True Positive (TP)**: The model correctly predicted a positive outcome (the actual outcome was positive).
- **True Negative (TN)**: The model correctly predicted a negative outcome (the actual outcome was negative).
- **False Positive (FP)**: The model incorrectly predicted a positive outcome (the actual outcome was negative). Also known as a Type I error.
- **False Negative (FN)**: The model incorrectly predicted a negative outcome (the actual outcome was positive). Also known as a Type II error.

| | **Predicted Positive** | **Predicted Negative** |
|---|---|---|
| **Actual Positive** | True Positive (TP) | False Negative (FN) |
| **Actual Negative** | False Positive (FP) | True Negative (TN) |

Table 1: Confusion Matrix

### 6.1. Classification report :

Classification report is one of the performance evaluation metrics of a classification-based machine learning model. It displays your model's precision, recall, F1 score and support. It provides a better understanding of the overall performance of our trained model.

- **Accuracy:** Accuracy is used to measure the performance of the model. It is the ratio of the sum of true positives and true negatives to the total number of instances (sum of all possible outcomes).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

- **Precision:** Precision is a measure of how accurate a model's positive predictions are. It is defined as the ratio of true positive predictions to the total number of positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

- **Recall:** Recall measures the effectiveness of a classification model in identifying all relevant instances from a dataset. It is the ratio of the number of true positive (TP) instances to the sum of true positive and false negative (FN) instances.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

- **F1-Score:** F1-score is used to evaluate the overall performance of a classification model. It is the harmonic mean of precision and recall.

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

### 6.2. Product Sentiment Classification Result

| Model | Type | Precision | Recall | F1 Score |
|---|---|---|---|---|
| CNN | Product | 0.93 | 0.80 | 0.86 |
| | Delivery | 0.82 | 0.94 | 0.88 |
| RCNN | Product | 0.94 | 0.79 | 0.86 |
| | Delivery | 0.82 | 0.95 | 0.88 |

Table 2: Model Performance Metrics of Classification for CNN and RCNN

**For CNN classification,** we are getting accuracy of 93% (Pression), 80% (Recal), 86% (F1 Score) For Product. And for Delivery We are getting accuracy of 82%(Pression), 94% (Recall), 88% (F1 Score).
**For RCNN classification,** we are getting accuracy of 94% (Pression), 79% (Recal), 86% (F1 Score) For Product. And for Delivery We are getting accuracy of 82%(Pression), 95% (Recall), 88% (F1 Score).

**For CNN Sentiment Analysis,** we are getting accuracy of 91% (Pression), 88% (Recal), 89% (F1 Score) for positive. And for negative, we are getting accuracy of 88%(Pression), 91% (Recall), 90% (F1 Score).
**For RCNN Sentiment Analysis ,** we are getting accuracy of 92% (Pression), 90% (Recal), 91% (F1 Score) for Positive. And for Negative we are getting accuracy of 90%(Pression), 92% (Recall),91% (F1 Score).

| Model | Type | Precision | Recall | F1 Score |
|---|---|---|---|---|
| CNN | Positive | 0.91 | 0.88 | 0.89 |
| | Negative | 0.88 | 0.91 | 0.90 |
| RCNN | Positive | 0.92 | 0.90 | 0.91 |
| | Negative | 0.90 | 0.92 | 0.91 |

Table 3: Model Performance Metrics of Sentiment for CNN and RCNN

### 6.3. CNN & R-CNN Classification Comparison:

For CNN classification, we are getting 87% Accuracy. In Confusion Matrix, for 100 of each category values we are getting about 80 true positives, 20 false positives and 94 true negative 6 false negatives values.
For RCNN Classification we are getting 90% Accuracy. In Confusion Matrix, for 100 of each category values we are getting about 78 true positives, 22 false positives and 96 true negative 4 false negatives values. Confusion Matrix.

We can see that, R-CNN is performing better than CNN in classification .

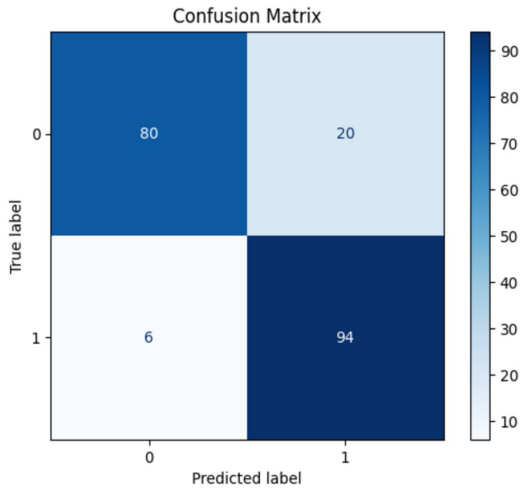



Figure 8: CNN Classification Confusion Matrix.

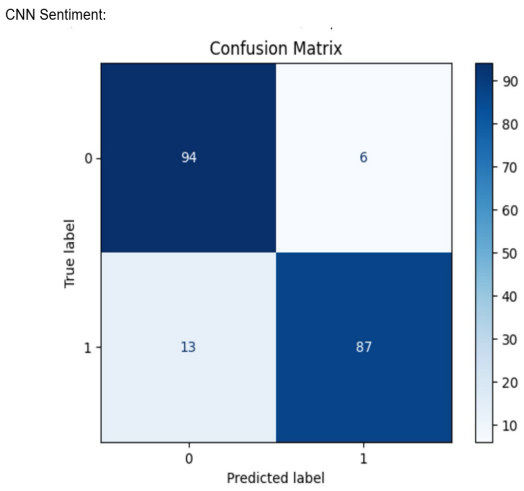



Figure 9: R-CNN Classification Confusion Matrix.

### 6.4. CNN & R-CNN Sentiment Analysis Comparison :

For Sentiment Analysis in CNN our accuracy is 91%. In Confusion Matrix, for 100 of each category values we are getting about 94 true positives, 6 false positives and 87 true negative 13 false negatives values.
For Sentiment Analysis in RCNN we are getting 90% Accuracy.In Confusion Matrix, for 100 of each category values we are getting about 91 true positives, 9 false positives and 89 true negative 11 false negatives values.

We can see that, CNN is performing slightly better than R-CNN in Sentiment Analysis .

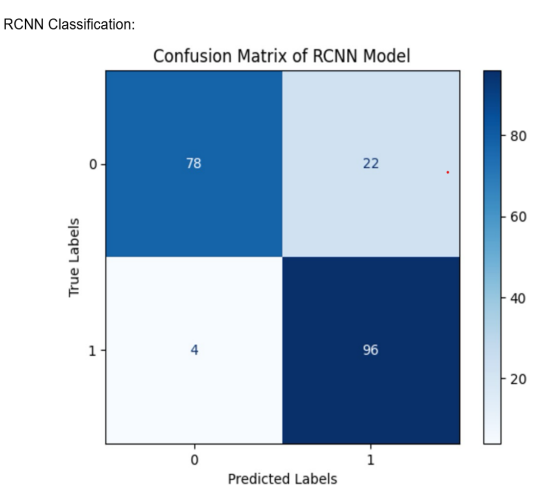



Figure 10: CNN Classification Confusion Matrix.

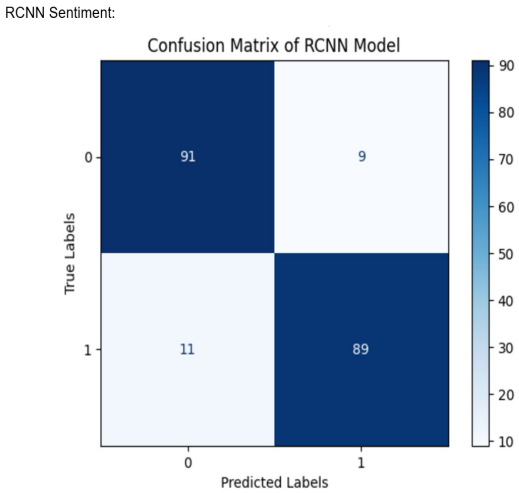



Figure 11: R-CNN Sentiment Analysis Confusion Matrix.

# CHAPTER 7

## 7. Limitations and Future Work

This study opens up future research directions that would help overcome challenges faced in the study and extend scope and depth of Bangla e-commerce review sentiment analysis . The directions for future investigation are suggested as follows:

### 7.1. Expansion of Dataset

The review dataset was limited in terms of diversity as it was based on a some limited web platform. At least some representativeness would be achieved by expanding it to reviews from different e-commerce platforms and more diverse customer demographics. This extension would enhance the generalizability of study results and enable cross-comparative studies among e-commerce environments.

### 7.2. Incorporating Mixed Sentences

Any review that presented mixed sentiments — mentioning both the product and delivery in the same breath — was excluded from this research. Future work could fill this gap, developing methods to parse and analyze such sentences containing more complex structures. It would offer a more detailed view of customer feedback and enable a broader sentiment analysis structure

### 7.3. Multilingual Analysis

The current dataset only contains Bangla reviews, but in the future multilingual sentiment analysis can be done considering the mixture of languages that is often present in Bangladeshi customers reviews due to the extreme code-switching. Sentiment analysis systems will be more robust and appealing to a wider audience if techniques to analyze Bangla-English mixed reviews were incorporated.

### 7.4. Aspect-Based Sentiment Analysis (ABSA)

A possible future approach to obtaining deeper insights on customer thoughts would be to adopt Aspect-Based Sentiment Analysis (ABSA). ABSA methods are able to identify sentiments related to particular aspects like product quality, delivery time, packing or customer service. It will help firms to pinpoint the areas where they need to make improvements and where they can focus on greater customer satisfaction.

### 7.5. Real-Time Sentiment Analysis

Another promising research direction for the future is designing real-time sentiment analysis systems for Bangla reviews. This kind of system would open the door for businesses to recognize and respond to customer complaints in an efficient manner to resolve the work issue

at a faster pace or improve the customer experience. Real-time Bangla sentiment analysis on a scalable system would help the e-commerce websites in Bangladesh tremendously.

### 7.6. Reduction of Dataset Bias

Our Dataset was biased due to uneven class distribution, platform specific feedback patterns, or over representation of certain types of sentiments. To alleviate this, possible future work might include the utilization of advanced techniques such as synthetic data augmentation, oversampling of minority classes or adversarial training . Moreover, more diverse data sources with balanced representation across sentiments and categories (such as products against delivery) will help improve model robustness and generalizability.

### 7.7. Use of Advanced NLP Architectures

Utilizing state of the art Bangla language models like BanglaBERT will greatly enhance performance in classifying sentiment. BanglaBERT is a transformer-based model and as it is pre-trained on a larger corpus of Bangla text, it can easily overcome complexities of Bangla language such as mixed-script reviews. Fine-tuning these models over the curated dataset would provide the models the ability to capture the linguistic nuances of the domain like code-switching and shifts in sentiment polarity within reviews. Open for future works transfer learning approaches can also be explored on other pre-trained multilingual models (e.g. XLM-Roberta and other pre-trained models) and tune that to Bangla specific tasks.

# CHAPTER 8

## 8. Conclusion

The purpose of this research is to build a dataset of Bangla customer reviews from e-commerce platforms and observe how Deep Learning models performed on it. In doing so, we effectively showed how CNN and RCNN can categorize reviews and sentiment analysis using relatively less data. This study contributes to the development and application of datasets for Bangla text as well as revealing that deep learning models can explore customer feedback data useful, which can be helpful to companies. The work has taken the first steps towards more complex approaches and a larger datasets as future work to achieve better results.

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